Predictive Modeling of Angular Insertion Depth in Lateral Wall Cochlear Implant Electrodes

Report submitted to Meharry Medical College
for the award of the degree

of

Masters of Science

by

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under the guidance of

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Certificate of Capstone Project Completion

This is to certify that the report entitled **Predictive Modeling of Angular Insertion Depth in Lateral Wall Cochlear Implant Electrodes**, submitted by **Michella Maddox-McGhee** to Meharry Medical College, is a record of bona fide research and practical work carried out under our supervision. The work presented in this report meets the academic and professional standards required for the successful completion of the Capstone Project as part of the Master of Data Science program at the Institute. In recognition of their dedication and scholarly contribution, this report is hereby deemed worthy of consideration for fulfillment of the degree requirements.

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- To my parents -

Dear Mom,

I wish I could have made you proud while your presence was still here with me on this Earth. There are so many moments I imagine sharing with you, especially this one, where I have reached a milestone that I know you would have cherished deeply. Even though you are no longer here in the physical sense, I carry your love, your lessons, and your spirit with me every single day.

I pray that your spiritual transcendence continues to guide me, instilling in me the faith and resilience I need to keep moving forward. In times of doubt or exhaustion, I lean on the strength I believe you have passed down to me. Everything I accomplish is, in part, a reflection of you and the foundation you helped build within me. I hope, wherever you are, you can see me and feel proud of the person I am becoming.

Dear Dad,

Thank you for giving me life and for believing in me in ways that have shaped my confidence and determination. Your support, whether spoken or unspoken, has meant more to me than I can fully express. You have played a vital role in helping me become who I am today, and I am deeply grateful for that.

Through every challenge and every success, I have felt the impact of your belief in me pushing me to keep going. I hope to continue making you proud as I grow, achieve, and pursue my goals. I love you more than words can fully capture, and I carry that love with me in everything I do.

Declaration

I certify that:

a. the work contained in the report is original and has been done by me under the guidance of my supervisor;

b. the work has not been submitted to any other institute for any other degree or diploma;

c. I have followed the guidelines provided by the Institute in preparing the thesis;

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e. whenever I have used materials (data, models, figures and text) from other sources, I have given due credit to them by citing them in the text of the thesis, and giving their details in the references, and taken permission from the copyright owners of the sources, whenever necessary.

Acknowledged By Michella Maddox-McGhee

Acknowledgement

I am grateful to my mentor **Dr. Mohammad Mahmudur Rahman Khan** for his continuous guidance, support, and mentorship throughout this project. His expertise and insight were instrumental in shaping this research.

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Executive Summary

This study focuses on predicting angular insertion depth in cochlear implant procedures using a machine learning approach. Accurate prediction of insertion depth is critical for optimizing cochlear coverage while preserving residual hearing.

The research utilizes CT imaging data from 86 cochlear implant patients and analyzes key anatomical and device related features, including Base Insertion Depth (BID), Angular Insertion Depth (AID), cochlear scale, electrode array length, and electrode diameter. Among these, BID represents the distance from the round window to the most basal electrode and serves as a primary indicator of insertion progression.

A Support Vector Machine (SVM) model was developed as the primary contribution of this study. The model achieved a mean absolute error of 27 degrees, a standard deviation of 36 degrees, and an absolute deviation error of 24 degrees.

These results were compared to prior research methods, including Linear Regression, which reported a standard deviation of 41 degrees and an absolute deviation error of 32 degrees. The comparison demonstrates that the SVM model provides improved predictive accuracy.

This work highlights the effectiveness of machine learning in surgical planning and emphasizes the importance of BID as a key predictive feature in improving patient outcomes.

Keywords: Cochlear Implants, BID (Base Insertion Depth), AID (Angular Insertion Depth), Machine Learning, SVM (Support Vector Machine)

Contents

List of Abbreviations

AID	Angular Insertion Depth
BID	Base Insertion Depth
SVM	Support Vector Machine
MAE	Mean Average Error
LOOCV	Leave-One-Out Cross-Validation
GridSearch CV	GridSearch Cross-Validation
AdaBoost	Adaptive Boost
XGBoost	eXtreme Gradient Boosting
KNN	K-Nearest Neighbor (K is a constant)
CT-based	Computed Tomography-based
A Len.	A Length
B. Len.	B Length
Pred. AID	Predicted Angular Insertion Depth
Arr. Len.	Array Length
Arr. Diam.	Array Diameter
MED-EL	Medical Electronics
et. al.	Latin abbreviation for "and others"
mm	Millimeter
pp	Pages
vol.	Volume
no.	Number
3d	3-dimensional

List of Tables

List of Figures

Project Narrative

1 Introduction

Hearing loss affects millions of individuals worldwide, and cochlear implants provide an effective treatment for severe to profound sensorineural hearing loss. These devices bypass damaged hair cells and directly stimulate the auditory nerve.

The placement of the electrode array within the cochlea plays a critical role in determining hearing outcomes. Angular insertion depth directly affects cochlear coverage and speech perception. Prior studies have demonstrated that insertion depth is closely associated with both speech perception outcomes and the preservation of residual hearing, which highlights the importance of accurate electrode placement [1, 2]. Moreover, improper insertion can damage delicate structures and reduce residual hearing.

One of the most important measurable factors in determining insertion outcomes is Base Insertion Depth (BID), which represents how far the electrode array has progressed from the round window at the basal region of the cochlea.

Due to limited visualization and anatomical variability, predicting optimal insertion depth remains a significant challenge. This study addresses this issue by developing a predictive model using machine learning techniques. In this study, angular insertion depth is treated as a supervised regression problem in which preoperative anatomical and device-related measurements are used to estimate the final insertion outcome. Framing the problem in this way allows the investigation of whether structured numerical features can provide reliable guidance for surgical planning before implantation of the electrode array.
A reliable predictive model has potential clinical value because it may support patient-specific decision making, improve electrode selection, and reduce the likelihood of insertion strategies that result in excessive trauma or suboptimal cochlear coverage.
In addition to developing the final predictive model, this study compares multiple machine learning approaches to determine which method provides the most accurate estimation of angular insertion depth in a small biomedical dataset.

1.1 Background

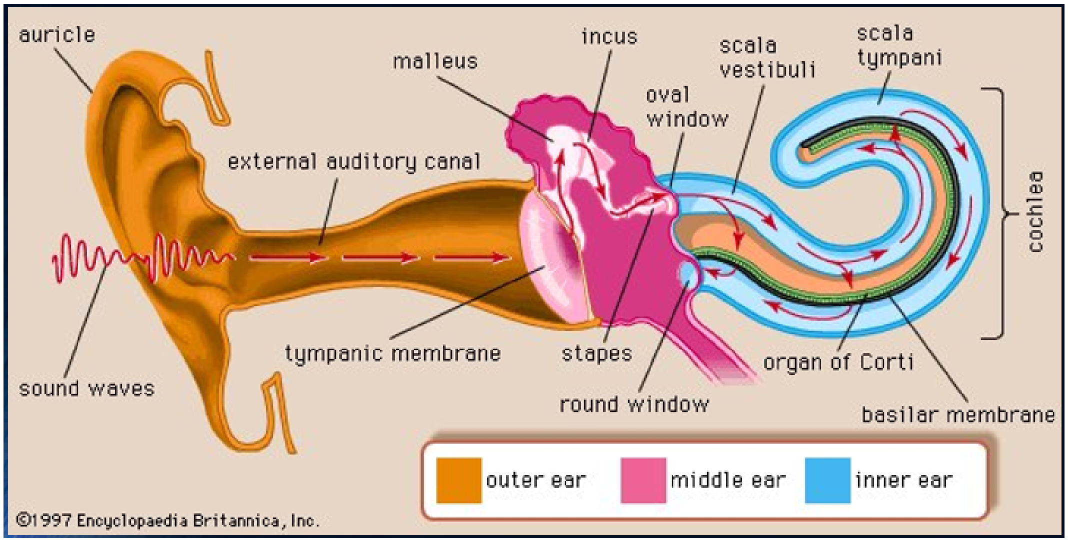

Figure 1: Ear Diagram

The cochlea is a spiral-shaped organ where different regions correspond to different sound frequencies. High-frequency sounds are detected near the base, while low-frequency sounds are detected deeper within the structure.

Cochlear implants are widely recognized as an effective treatment for individuals with severe to profound sensorineural hearing loss, providing direct stimulation of the auditory nerve when traditional hearing aids are insufficient [3].

Variability in cochlear anatomy makes standardized surgical approaches ineffective. Because cochlear morphology differs from patient to patient, the same electrode array may produce substantially different insertion outcomes across individuals. This variability makes patient-specific prediction especially important when attempting to balance deep cochlear coverage against the risk of damaging delicate intracochlear structures. Over-insertion of electrode arrays can damage functional regions responsible for residual hearing. It is said that manual insertion of the electrode array may result in damaging the soft tissue structures and basilar membrane. Surgeons are looking into using an automated insertion device because there are reports that it's less traumatic in cochlear implant surgery [4]. To address this challenge, robotic and automated insertion systems have been proposed to reduce variability and minimize insertion forces, thereby improving surgical precision [5–8].

Manual insertion is usually done with microforceps being held by a person. Intracochlear trauma during electrode insertion has been identified as a major contributor to postoperative hearing loss and reduced implant effectiveness [1, 9, 10].

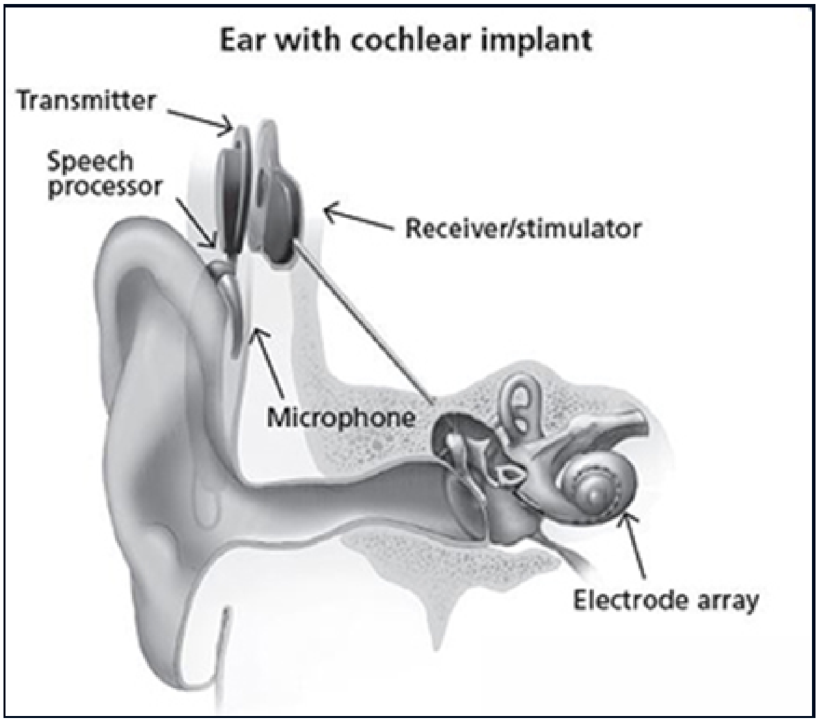

Figure 2: Ear Diagram w/ Electrode Array Implant

Measurements such as Base Insertion Depth, Cochlear Scale, and electrode dimensions provide a practical basis for predictive modeling because they capture both anatomical constraints and device-related characteristics that influence insertion behavior. When analyzed together, these variables offer a data-driven alternative to relying solely on generalized assumptions or fixed insertion strategies. Therefore, accurate prediction of insertion depth is essential for improving surgical outcomes.

2 Objectives

The primary objective of this study is to develop a predictive model for estimating angular insertion depth using anatomical and device-related features, with a particular emphasis on Base Insertion Depth.

2.1 Research Hypothesis/Questions

A Support Vector Machine model utilizing features such as Base Insertion Depth can more accurately predict angular insertion depth compared to traditional methods such as Linear Regression and geometric models.

2.2 Significance

This research contributes to improved surgical planning by providing a data-driven approach to electrode placement. Incorporating BID into predictive modeling allows for more precise estimation of insertion depth and reduces the risk of damaging residual hearing.

2.2.1 Organization of the Report

The remainder of this report is organized as follows. Section 3 presents the design and methodology, including the proposed Support Vector Machine model, the additional machine learning models evaluated, and the data processing and analysis techniques used in this study. This section also describes the dataset and outlines the feature relationships explored through correlation analysis.

Section 4 presents the results and discussion, including a comparative evaluation of all machine learning models based on Mean Absolute Error (MAE). This section further analyzes the relationships between variables and interprets the performance of the proposed model in the context of existing methods.

Section 5 provides the conclusions of the study, summarizing the key findings and contributions. Section 6 outlines future work and recommendations for extending this research. Section 7 reviews related work in cochlear implant modeling, imaging, and machine learning. Finally, the appendix includes supporting material such as feature definitions and model performance metrics.

3 Design and Methodology

This study follows a data-driven approach to model the relationship between anatomical features and angular insertion depth. The methodology focuses on extracting meaningful patterns from structured measurement data and applying machine learning techniques to capture both linear and non-linear relationships. Special emphasis is placed on understanding how Base Insertion Depth interacts with other features to influence outcomes.

3.1 Proposed Method

A Support Vector Machine (SVM) model with a linear kernel was implemented as the best predictive model in this study. The linear kernel was selected to evaluate whether a structured linear relationship exists between anatomical features and angular insertion depth, while still benefiting from the regularization and margin maximization properties of SVM.

Given the relatively small dataset size (n = 86), Leave-One-Out Cross-Validation (LOOCV) was used to ensure robust model evaluation. In this approach, each observation is used once as a test sample while the remaining data are used for training, minimizing bias and maximizing data utilization.

Feature scaling was applied prior to model training to normalize the input variables, ensuring that differences in measurement scales did not disproportionately influence the model.

The SVM model achieved a Mean Absolute Error (MAE) of 27.218774 degrees, demonstrating strong predictive performance relative to other evaluated machine learning models. Advanced computational methods have been applied in prior studies to model cochlear anatomy and electrode positioning using imaging data, demonstrating the potential of data-driven approaches for improving prediction accuracy [11–13]. Machine learning techniques have increasingly been applied in medical imaging and surgical planning to model complex anatomical relationships that are difficult to capture using traditional regression-based approaches [12, 13].

The SVM model was chosen due to its ability to capture non-linear relationships between variables, which are expected in biological and anatomical systems.

3.2 Other Machine Learning Models Evaluated

Gradient Boosting was evaluated as an ensemble learning approach for predicting angular insertion depth. This method builds decision trees sequentially, with each new tree correcting the residual errors of the previous ensemble. A GradientBoostingRegressor implementation from scikit-learn was used. Hyperparameters were optimized using GridSearchCV with Leave-One-Out Cross-Validation (LOOCV), and model selection was based on Mean Absolute Error (MAE). The search included variations in the number of estimators, learning rate, tree depth, minimum split size, minimum leaf size, subsampling, and feature selection strategy. The best Gradient Boosting model achieved an MAE of 33.454194 degrees.

AdaBoost was evaluated as an ensemble regression method that combines multiple weak learners to improve the accuracy of the prediction. In this study, a DecisionTreeRegressor was used as the base learner within an AdaBoostRegressor framework. Hyperparameters, including the number of estimators, learning rate, tree depth, and minimum leaf size, were explored using GridSearchCV. Final performance was then assessed using Leave-One-Out Cross-Validation (LOOCV) and Mean Absolute Error (MAE). The best AdaBoost configuration achieved an MAE of 36.761919 degrees.

K-Nearest Neighbors (KNN) regression was evaluated as a non-parametric method for predicting angular insertion depth. This approach estimates the target value based on the average of the nearest neighboring observations in the feature space. Feature scaling was applied using standardization to ensure that all variables contributed equally to distance calculations. Hyperparameters, including the number of neighbors, distance weighting scheme, and distance metric, were optimized using GridSearchCV. Final model performance was evaluated using Leave-One-Out Cross-Validation (LOOCV) with Mean

Absolute Error (MAE). The best KNN model achieved an MAE of 35.997671 degrees.

Random Forest regression was evaluated as an ensemble learning method that constructs multiple decision trees using bootstrap sampling and aggregates their predictions to reduce variance. A RandomForestRegressor implementation from scikit-learn was used. Hyperparameters, including the number of trees, tree depth, minimum split size, minimum leaf size, and feature selection strategy, were optimized using GridSearchCV. Final model performance was evaluated using Leave-One-Out Cross-Validation (LOOCV) with Mean Absolute Error (MAE). The best Random Forest model achieved an MAE of 38.519750 degrees.

Extreme Gradient Boosting (XGBoost) was evaluated as an advanced ensemble learning method that builds decision trees sequentially while optimizing a regularized objective function. The XGBRegressor implementation was used, and hyperparameters such as the number of estimators, learning rate, maximum tree depth, subsampling ratio, and feature sampling ratio were tuned using GridSearchCV. Final model performance was assessed using Leave-One-Out Cross-Validation (LOOCV) with Mean Absolute Error (MAE). The best XGBoost model achieved an MAE of 38.4601487 degrees.

3.3 Tools and Technologies

Python was used as the primary programming language for data processing, analysis, and model development. Machine learning models were implemented using the Scikit-learn library, which provided tools for regression modeling, hyperparameter tuning, and performance evaluation.
For ensemble-based models, additional libraries such as XGBoost were utilized to implement gradient boosting techniques with optimized performance. Data preprocessing steps, including feature scaling and transformation, were performed to ensure consistency across model inputs and to improve model stability.

Model evaluation was conducted using Leave-One-Out Cross-Validation (LOOCV), which is particularly well-suited for small datasets. This approach maximizes the use of available data by iteratively training on all but one observation and testing on the remaining sample.

Additionally, computational tools were used to support debugging and code refinement during development, improving the efficiency and reliability of the modeling workflow.

3.4 Data Gathering

The dataset used in this study was obtained from prior research conducted by Dr. Mohammad Mahmudur Rahman Khan.

The data consisted exclusively of 86 numerical measurements extracted from CT scans, rather than raw imaging data. These measurements included Base Insertion Depth, A

Length, B Length, Cochlear Scale, Electrode Array Length, Electrode Array Diameter, Actual Angular Insertion Depth, and Predicted Angular Insertion Depth. A range of quantitative measurements derived from CT imaging have been widely used in prior research to estimate cochlear geometry and electrode positioning with high precision [11, 13].

3.5 Complete Dataset

Table 2: The Complete Dataset Used for Model Development and Evaluation

BID	Scale	A Len.	B Len.	Type	Actual AID	Pred. AID	Arr. Len.	Arr. Diam.
-0.32	2.67	8.67	6.38	CI522	342	362	23	0.4
2.25	2.55	8.23	6.11	CI522	473	452	23	0.4
3	2.63	8.6	6.3	CI522	413	453	23	0.4
2.3	2.86	9.41	6.84	CI522	371	375	23	0.4
4.1	3.04	9.92	7.3	CI522	404	369	23	0.4
1.25	2.75	9.02	6.47	CI522	385	378	23	0.4
1.1	2.94	9.53	7.15	CI522	328	327	23	0.4
4.7	2.77	9.03	6.6	CI522	422	459	23	0.4
3.45	2.82	9.27	6.76	CI522	389	415	23	0.4
1.1	2.77	8.9	6.77	CI522	363	369	23	0.4
0.1	2.76	8.94	6.58	CI522	352	349	23	0.4
4.75	2.88	9.33	6.94	CI522	395	432	23	0.4
2.65	2.95	9.61	7.04	CI522	357	362	23	0.4
1.75	2.84	9.32	6.73	CI522	409	365	23	0.4
-0.57	2.83	9.16	6.71	CI522	359	310	23	0.4
3.6	2.82	9.3	6.72	CI522	422	416	23	0.4
3.65	2.85	9.25	6.76	Flex24	461	456	24	0.5
2.25	2.83	9.13	6.66	Flex28	581	524	28	0.6
2.45	2.7	8.88	6.49	Flex28	637	560	28	0.6
3.25	2.83	9.18	6.88	Flex28	489	551	28	0.6
3.65	2.94	9.55	6.96	Flex28	513	533	28	0.6
3.05	2.97	9.61	7.13	Flex28	449	512	28	0.6
3.35	2.95	9.47	7.11	Flex28	465	523	28	0.6
2.35	2.8	9.04	6.57	Flex28	545	536	28	0.6
2.1	2.73	8.89	6.51	Flex28	584	547	28	0.6
2.15	2.9	9.47	6.91	Flex28	454	508	28	0.6
2.85	2.71	8.86	6.43	Flex28	637	568	28	0.6
1.7	2.58	8.45	6.2	Flex28	600	574	28	0.6
3	2.82	9.08	6.79	Flex28	537	546	28	0.6
-0.68	2.67	8.66	6.32	Flex28	500	495	28	0.6
3.2	2.53	8.16	6.11	Flex28	717	614	28	0.6
1.85	2.76	8.99	6.62	Flex28	516	536	28	0.6
3.25	2.75	8.86	6.56	Flex28	510	573	28	0.6
1.85	2.76	8.93	6.66	Flex28	483	535	28	0.6
1.4	2.81	9.08	6.67	Flex28	584	510	28	0.6
2	2.73	8.85	6.57	Flex28	523	547	28	0.6
0.7	2.58	8.48	6.08	Flex28	513	554	28	0.6

Continued on next page

BID	Scale	A Len.	B Len.	Type	Actual AID	Pred. AID	Arr. Len.	Arr. Diam.
1.9	2.99	9.74	7.24	Flex28	502	475	28	0.6
-2.06	2.77	8.89	6.58	Flex28	416	442	28	0.6
-3.44	2.62	8.61	6.23	Flex28	394	450	28	0.6
-1.33	2.74	8.95	6.53	Flex28	478	461	28	0.6
0.1	2.77	9.03	6.67	Flex28	474	491	28	0.6
3.35	3.06	9.96	7.24	Flex28	527	492	28	0.6
1.8	2.77	8.97	6.48	Flex28	554	529	28	0.6
-0.85	2.85	9.2	6.84	Flex28	426	449	28	0.6
1.9	2.98	9.47	7.25	Flex28	458	480	28	0.6
1.75	2.73	9.01	6.52	Flex28	603	537	28	0.6
4.4	2.81	9.09	6.75	Flex28	581	583	28	0.6
2.5	2.76	8.85	6.69	HiFocus SlimJ	472	448	23	0.55
-0.98	2.71	8.9	6.45	HiFocus SlimJ	366	379	23	0.55
1.75	2.81	9.19	6.77	HiFocus SlimJ	398	419	23	0.55
-0.2	2.97	9.65	7.1	HiFocus SlimJ	357	330	23	0.55
1.55	2.66	8.51	6.29	HiFocus SlimJ	480	448	23	0.55
3.17	2.79	9	6.82	HiFocus SlimJ	422	458	23	0.55
3.6	2.74	8.91	6.47	HiFocus SlimJ	419	483	23	0.55
0.7	2.79	8.96	6.76	HiFocus SlimJ	376	398	23	0.55
2.35	3.01	9.92	6.99	HiFocus SlimJ	374	383	23	0.55
-0.64	2.78	9	6.71	HiFocus SlimJ	336	370	23	0.55
4.69	2.81	9.15	6.71	HiFocus SlimJ	467	490	23	0.55
2.6	2.8	9.07	6.61	HiFocus SlimJ	432	442	23	0.55
2.85	2.94	9.57	7.02	HiFocus SlimJ	443	409	23	0.55
1.45	2.69	8.43	6.36	HiFocus SlimJ	450	441	23	0.55
-0.28	2.71	8.74	6.57	HiFocus SlimJ	391	396	23	0.55
-0.39	2.98	9.63	7.21	HiFocus SlimJ	311	325	23	0.55
-1.95	2.95	9.56	7.17	HiFocus SlimJ	398	280	23	0.55
1.1	2.64	8.52	6.36	HiFocus SlimJ	402	449	23	0.55
2.05	2.67	8.69	6.41	HiFocus SlimJ	538	456	23	0.55
0.55	2.68	8.66	6.47	HiFocus SlimJ	406	422	23	0.55
3.54	2.74	8.98	6.52	HiFocus SlimJ	487	477	23	0.55
2	2.88	9.05	6.88	HiFocus SlimJ	416	407	23	0.55
2.35	3.07	9.94	7.25	HiFocus SlimJ	365	366	23	0.55
2.7	2.85	9.29	6.78	HiFocus SlimJ	394	431	23	0.55
-0.12	2.92	9.56	6.96	HiFocus SlimJ	340	345	23	0.55
2.05	2.72	8.91	6.47	HiFocus SlimJ	431	449	23	0.55
-0.96	2.68	8.74	6.37	HiFocus SlimJ	374	387	23	0.55
4.57	3.09	9.94	7.37	HiFocus SlimJ	394	417	23	0.55
2.75	2.85	9.24	6.67	HiFocus SlimJ	417	433	23	0.55
2.9	3.03	9.97	7.13	HiFocus SlimJ	455	386	23	0.55
1.2	2.84	9.17	6.82	HiFocus SlimJ	388	398	23	0.55
2.25	2.9	9.44	6.96	HiFocus SlimJ	475	404	23	0.55
1.4	2.79	9.03	6.74	HiFocus SlimJ	371	415	23	0.55
0.75	2.76	8.97	6.79	Standard	637	610	31.5	0.75
1.8	2.78	8.9	6.71	Standard	661	629	31.5	0.75
3.2	3.08	9.98	7.28	Standard	584	591	31.5	0.75
0.45	2.91	9.48	7.09	Standard	584	566	31.5	0.75
-2.63	2.8	9.28	6.72	Standard	447	532	31.5	0.75

No image processing was performed in this study, as the dataset was already preprocessed into a structured numerical form.

Electrode Array Type	**Number of Cases**	**Array Length (in mm)**	**Array Diameter (in mm)**
MED-EL (Standard)	5	31.5	0.75
MED-EL (Flex28)	31	28	0.6
MED-EL (Flex24)	1	24	0.5
Cochlear (CI522)	16	23	0.4
Advanced Bionics (HiFocus SlimJ)	33	23	0.55

Table 3: Compact Data Table

3.6 Data Analysis

Correlation and partial correlation analyses were conducted to evaluate the relationships between the predictor variables and angular insertion depth. These analyses were used to identify both the strength of association and the independence of each feature in contributing to the prediction task.

The initial correlation matrix provided insight into pairwise relationships between variables, revealing which features were most strongly associated with angular insertion depth. In particular, Base Insertion Depth (BID) demonstrated a consistent positive relationship with the target variable, suggesting its importance as a primary predictor.

To further isolate the influence of individual variables, partial correlation analysis was performed. This approach allowed for the examination of relationships while controlling for the effects of other features, such as cochlear scale and electrode dimensions. The results confirmed that BID maintained a strong independent association with angular insertion depth, reinforcing its significance in predictive modeling.

These analytical steps provided a data-driven foundation for model development, guiding feature selection and supporting the use of machine learning techniques to capture both direct and interacting effects among variables.

4 Results and Discussions

The results of this study highlight strong relationships between several key variables and angular insertion depth, with Base Insertion Depth (BID) emerging as one of the most influential predictors.

Initial correlation analysis showed a strong positive relationship between BID and angular insertion depth. As BID increased, angular insertion depth consistently increased, indicating that BID serves as a direct indicator of electrode progression within the cochlea. This trend aligns with previous findings indicating that electrode advancement is strongly correlated with insertion depth metrics obtained from imaging and electrophysiological measurements [14, 15].

4.1 Model Performance Comparison

To evaluate model suitability for predicting angular insertion depth, six machine learning approaches were compared using Mean Absolute Error (MAE). The models included AdaBoost, Gradient Boosting, K-Nearest Neighbors, Random Forest, XGBoost, and Support Vector Machine. As shown in Table 4, the linear SVM achieved the lowest MAE, indicating the strongest predictive performance among the evaluated methods.

Model	**MAE (degrees)**
AdaBoost	36.761919
Gradient Boosting	33.454194
K-Nearest Neighbors (KNN)	35.997671
Random Forest	38.519750
XGBoost	38.4601487
Support Vector Machine	**27.218774**

Table 4: A Comparison of Machine Learning Model Performance

The comparative results indicate that the linear Support Vector Machine model outperformed all other evaluated approaches by a substantial margin. Although ensemble tree-based methods such as Gradient Boosting, AdaBoost, Random Forest, and XGBoost are often effective for nonlinear prediction tasks, their performance was weaker in this study. This pattern suggests that, for the present dataset, a regularized linear model generalized more effectively than more complex ensemble methods. The findings also indicate that the relationship between the predictor variables and angular insertion depth is structured enough to be captured by SVM without requiring highly flexible tree-based models.

Partial correlation analysis further confirmed this relationship by controlling for other variables such as cochlear scale and electrode dimensions. Even after accounting for these factors, BID maintained a strong independent effect, reinforcing its importance in predictive modeling.

Cochlear scale also demonstrated a meaningful relationship with angular insertion depth. Variability in cochlear size contributed to differences in insertion outcomes, highlighting the need for patient-specific modeling.

	Angular Insertion depth	BID	Cochlear Scale	Electrode Array Length	Electrode Array Diameter
Angular Insertion depth	1	0.2735	-0.2557	0.7431	0.6172
BID	0.2735	1	0.2510	-0.1037	-0.1769
Cochlear Scale	-0.2557	0.2510	1	-0.0400	0.0330
Electrode Array Length	0.7431	-0.1037	-0.0400	1	0.7661
Electrode Array Diameter	0.6172	-0.1769	0.0330	0.7661	1

Table 5: Correlation Matrix

	Angular Insertion depth	BID	Cochlear Scale	Electrode Array Length	Electrode Array Diameter
Angular Insertion depth	1	0.7190	-0.625	0.637	0.3805

Table 6: Partial Correlation Matrix

Electrode array length showed a positive correlation with angular insertion depth, as longer arrays are capable of deeper insertion. Electrode diameter, while less influential, introduced variability due to mechanical resistance during insertion.

The Support Vector Machine model achieved a mean absolute error of 27 degrees, a standard deviation of 36 degrees, and an absolute deviation error of 24 degrees. Traditional approaches, such as geometric modeling and linear regression, have reported higher variability in predicting insertion depth, underscoring the limitations of these methods in capturing patient-specific anatomical variability [11]. Overall, these results indicate strong predictive performance.

Compared to prior research using Linear Regression, which reported a standard deviation of 41 degrees and an absolute deviation error of 32 degrees, the SVM model demonstrated improved accuracy.

The improved performance of the SVM model suggests that the relationships between variables are not purely linear. Instead, the model captures more complex interactions, particularly involving Base Insertion Depth and anatomical variability.

5 Conclusions

This study demonstrates that angular insertion depth can be effectively predicted using machine learning techniques applied to structured anatomical and device-related measurements. Among the evaluated models, the linear Support Vector Machine (SVM) achieved the best performance, with the lowest Mean Absolute Error (MAE) compared to all other approaches.

The results highlight the importance of Base Insertion Depth (BID) as a key predictor, consistently showing a strong relationship with angular insertion depth across both correlation analysis and predictive modeling. These findings support the use of BID as a critical feature in preoperative planning.

A comparative evaluation of multiple machine learning models revealed that more complex ensemble methods, such as Random Forest and XGBoost, did not outperform the SVM in this study. This outcome suggests that, for small biomedical datasets, simpler and well-regularized models may generalize more effectively than highly flexible approaches that are prone to overfitting.

Overall, this research demonstrates the potential of data-driven methods to improve surgical planning in cochlear implantation. By providing accurate predictions of insertion depth, the proposed approach may contribute to better preservation of residual hearing and more personalized treatment strategies. These findings also emphasize the importance of selecting appropriate modeling techniques based on dataset size and structure.

6 Future Scopes/Recommendations

Future work should focus on expanding the dataset to include a larger and more diverse patient population. This would improve model generalizability and robustness.

Additional features could also be incorporated, such as more detailed anatomical measurements or patient-specific clinical variables.

Exploring other machine learning models, including ensemble methods or deep learning approaches, may further improve prediction accuracy.

Finally, integration of the model into clinical decision support systems could provide real-time guidance to surgeons, enhancing surgical precision and patient outcomes. The outcome of the surgery and life thereafter could be explored and documented for the sole purpose of improving the lives of future patients.

7 Related Works

Previous studies have investigated the relationship between electrode insertion depth and hearing outcomes. Many of these studies rely on geometric models or Linear Regression approaches, which assume simplified relationships between variables.

While these methods provide a foundation, they often fail to capture the complexity of cochlear anatomy and variability across patients.

Recent research has begun to explore data-driven approaches, but limited work has focused specifically on combining Base Insertion Depth with machine learning techniques.

This study builds upon prior research by incorporating SVM modeling and emphasizing the importance of BID as a primary predictor, offering improved accuracy and clinical relevance.

Several studies have investigated cochlear implant electrode insertion and its impact on hearing outcomes. Early research focused on understanding the relationship between insertion depth and intracochlear trauma, demonstrating that deeper insertions can increase the risk of structural damage [1].

In addition, systematic reviews of cochlear implant outcomes have highlighted the importance of preserving residual hearing and understanding electrophysiological responses in patients with cochlear implants [16].

Advancements in surgical techniques have led to the development of robotic and automated insertion systems aimed at reducing trauma and improving consistency during implantation [5, 6]. These approaches emphasize the importance of controlled insertion forces and precise alignment with cochlear anatomy.

In retrospect, imaging-based methods have been developed to improve the localization and measurement of cochlear structures. Techniques such as graph-based electrode localization and CT-based anatomical modeling have enabled more accurate estimation of electrode positioning [13, 17].

Despite these advancements, many existing approaches rely on simplified geometric assumptions or traditional regression models, which may not fully capture the complexity of anatomical variability. This study extends prior work by applying machine learning techniques that demonstrate that a Support Vector Machine model can improve predictive accuracy for angular insertion depth in a small biomedical dataset.

8 Appendix

Appendix A: Feature Definitions

- **Base Insertion Depth (BID):** Distance from the round window to the most basal electrode
- **Angular Insertion Depth:** Angle representing electrode position within the cochlea
- **Cochlear Scale:** Measure of cochlear size
- **Electrode Length:** Length of the electrode array
- **Electrode Diameter:** Thickness of the electrode

Appendix B: Model Metrics

- **SVM Mean Absolute Error:** 27 degrees
- **SVM Standard Deviation:** 36 degrees
- **SVM Absolute Deviation Error:** 24 degrees

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